

Assignment

module -5 Network Fundamentals and Building Networks

Section 1: Multiple Choice

1. What is the primary function of a router in a computer network?

- (a) Assigning IP addresses to devices
- (b) Providing wireless connectivity to devices
- (c) Forwarding data packets between networks
- (d) Managing user authentication and access control

Ans: (c) Forwarding data packets between networks

2. What is the purpose of DHCP (Dynamic Host Configuration Protocol) in a computer network?

- (a) Assigning static IP addresses to devices
- (b) Resolving domain names to IP addresses
- (c) Managing network traffic and congestion
- (d) Dynamically assigning IP addresses to devices

Ans: (d) Dynamically assigning IP addresses to devices

3. Which network device operates at Layer 2 (Data Link Layer) of the OSI model and forwards data packets based on MAC addresses?

- (a) Router
- (b) Switch
- (c) Hub
- (d) Repeater

Ans: (b) Switch

4. Which network topology connects all devices in a linear fashion, with each device connected to a central cable or backbone?

- (a) Star
- (b) Bus
- (c) Ring
- (d) Mesh

Ans: (b) Bus

Section 2: True or False

5. A VLAN (Virtual Local Area Network) allows network administrators to logically segment a single physical network into multiple virtual networks, each with its own broadcast domain.

Ans: True

6. TCP (Transmission Control Protocol) is a connectionless protocol that provides reliable, ordered, and error-checked delivery of data packets over a network.

Ans: False

7. A firewall is a hardware or software-based security system that monitors and controls incoming and outgoing network traffic based on predetermined security rules.

Ans: True

Section 3: Shorts answer

8. Describe the steps involved in setting up a wireless network for a small office or home office (SOHO) environment.

Ans: (1) **Plan** – Determine coverage area and number of devices.

- (2) **Connect hardware** – Modem → Router → Power on.
- (3) **Access router** – Log in via browser (192. 168. x. x)
- (4) **Configure Wi-Fi** – Set SSID, strong password, WPA2/WPA3.
- (5) **Enable DHCP** – Auto-assign IPs.
- (6) **Test** – Connect devices, check internet.
- (7) **Secure** – Change router password, enable firewall, update firmware.

Section 4: Practical Application

9. Demonstrate how to configure a router for Internet access using DHCP (Dynamic Host Configuration Protocol).

Ans: (1) Connect modem → router, power on.

(2) Open browser, go to router IP (192. 168. 1. 1), log in.

(3) In WAN/Internet settings, Set Connection Type = DHCP/Automatic IP

(4) Ensure LAN DHCP is enabled for devices.

(5) Save settings and test Internet.

Section 5: Essay

10. Discuss the importance of network documentation in the context of building and managing networks.

Ans: Importance of Network Documentation:

- 1. **Troubleshooting Made Easy** – Clear documentation of devices, IP addresses, and connections helps quickly identify and resolve network issues.

2. **Efficient Network Management** – Keeps track of configurations, VLANs, routers, switches, and Wi-Fi setups, making upgrades or changes smoother.
3. **Security and Compliance** – Helps monitor network access, track vulnerabilities, and maintain compliance with organizational or regulatory standards.
4. **Disaster Recovery** – In case of network failure, documentation provides a reference for restoring systems quickly.
5. **Training and Knowledge Sharing** – New IT staff can understand the network setup faster using documented diagrams and notes.
6. **Planning and Expansion** – Helps plan capacity, add devices, or expand the network without conflicts.